Using plug-ins to insert technical figures

by Nicolas Ratier

Updated for TeXmacs 2.1.5 in 2026.

1. TeXmacs plug-ins
2. DraTeX/AlDraTeX
3. Gnuplot
4. Graphviz
5. Xy-pic

1 TeXmacs plug-ins

TeXmacs communicates with external programs through *plug-ins*. Each plug-in is a folder in the `plugins` directory of TeXmacs (on macOS, inside the application: `TeXmacs.app/Contents/Resources/share/TeXmacs/plugins`); you can add your own plug-ins in the folder `plugins` of your TeXmacs home folder (`~/.TeXmacs` on Linux and macOS, `%APPDATA%\TeXmacs` on Windows). A plug-in usually needs an external program (a computer algebra system, LaTeX, Gnuplot, ...): TeXmacs looks for it in the `PATH`, and only the plug-ins whose program was found appear in the menu Insert→Session. The result of this detection is kept in a cache, and is redone only when the folders of the `PATH` change. This menu starts a *session*, in which you type commands for the program and get its answers, formulas or pictures, right in your document.



Figure 1. The menu Insert→Session, with the plug-ins found on this computer.

If you install a program after TEXMACS, or if it is not in the PATH, use Insert→Session→Redetect to empty this cache and look for the programs again, and, on MACOS and WINDOWS, Insert→Session→Manual path to tell TEXMACS where a program is. Some plug-ins have options in Insert→Session→Preferences.

Among the plug-ins of TEXMACS 2.1.5, we find:

- Computer algebra systems

axiom, fricas	The AXIOM system and its successor FRICAS
cadabra	Field theory
cocoa5	Commutative algebra
giac	The computer algebra system of XCAS
gtybalt	Built on GINAC
macaulay2	Algebraic geometry and commutative algebra
maple	
mathemagix	
mathematica	
maxima	
mupad	
mycas	A tiny example plug-in for developers
pari	Number theory
reduce	
sage	
sympy	Symbolic mathematics in PYTHON
yacas	

- Numerical computation software

lush	
matlab	
octave	
python	
qcl	Simulation of quantum algorithms
r	Statistical computing
scilab	

- Drawing programs

asymptote	Vector graphics language
dratex	Drawing different types of charts and diagrams
eukleides	Euclidean geometry
feynmf	Feynman diagrams
gnuplot	Data and function plotting
graph	One session for several of these programs
graphviz	Automatic graph visualization
plantuml	UML diagrams
texgraph	Mathematical graphics
tikz	PGF/TIKZ pictures
xypic	Commutative diagrams

- Other plug-ins: shells (shell, powershell), programming languages (lisp, the proof assistant coq, jupyter kernels), and the AI plug-in (sessions with language models).

This tutorial is about four of the drawing programs (in red). Each of them turns a short text description into a picture, which the session inserts in the document as a POSTSCRIPT image: the picture is saved with the document and printed in vector quality. An EPS picture comes at 0.618 of the width of the paragraph (0.8 for GNUPLOT); see the section on GNUPLOT for changing it. In this document, the small pictures of DRATEX and XY-PIC were set back to their natural size.

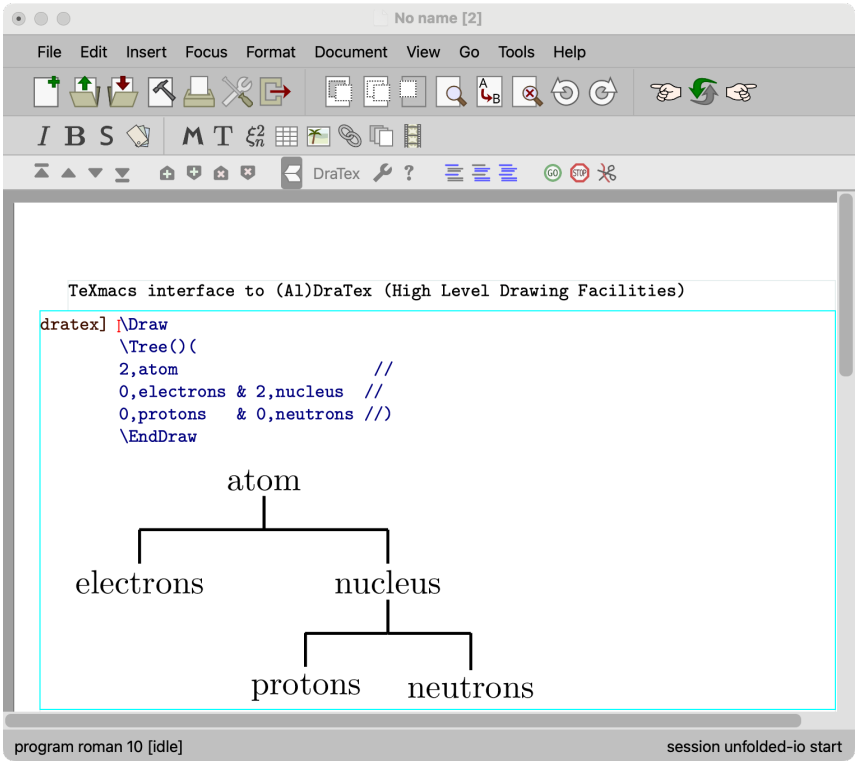


Figure 2. A DraTeX session in the window of TeXmacs. When the cursor is in the session, the focus toolbar shows the commands of the session: move between the fields, insert or remove fields, fold or unfold the field, the preferences of the plug-in, the help of the plug-in, evaluate, interrupt, close the session (the same commands are in the menu Focus).

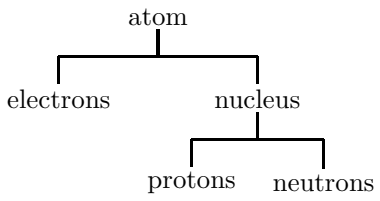
2 DraTeX/AlDraTeX

- DraTeX provides commands for drawing basic shapes like lines, rectangles, and Bézier curves, and for defining utilities that produce more complex outcomes.
- AlDraTeX provides templates for drawing different types of charts and diagrams.

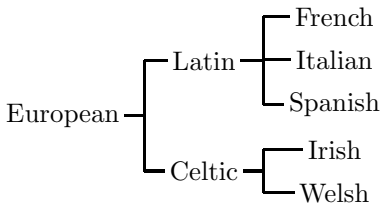
Start the session with Insert→Session→DraTex. It needs LaTeX and `dvips` (and Python, which runs the plug-ins of the drawing programs); the two style files `DraTex.sty` and `AlDraTex.sty` come with the plug-in. Type the drawing between `\Draw` and `\EndDraw`; ⇧↵ starts a new line in the input, and ↵ sends the input to LaTeX.

```
TeXmacs interface to (Al)DraTex (High Level Drawing Facilities)
```

```
dratex] \Draw
        \Tree()(
        2,atom                    //
        0,electrons & 2,nucleus   //
        0,protons   & 0,neutrons //)
        \EndDraw
```



```
dratex] \Draw
        \TreeAlign(H,0,0)(0,0,0)
        \TreeSpace(S,5,15)
        \Tree()(
        2,European //
        3,Latin  & 2,Celtic //
        0,French & 0,Italian & 0,Spanish & 0,Irish & 0,Welsh //)
        \EndDraw
```



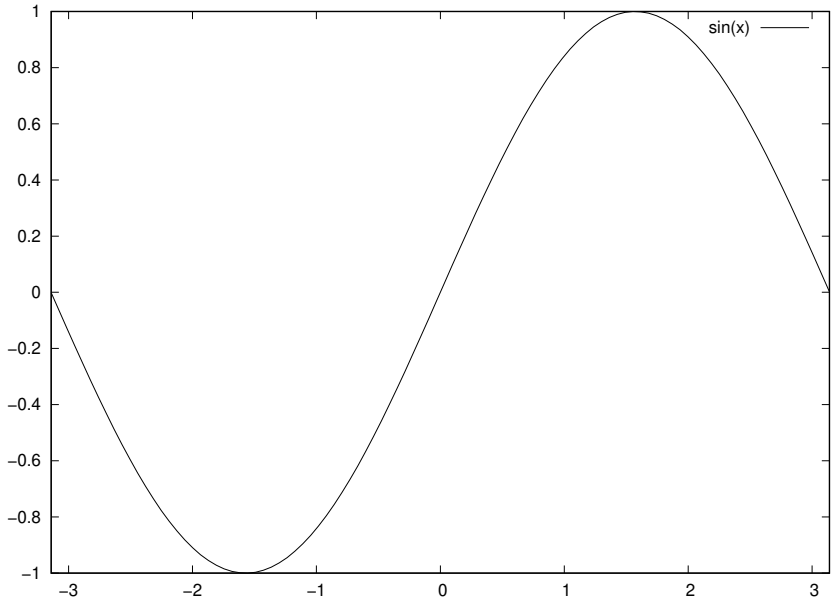

```
dratex]
```

3 GNUPLOT

- GNUPLOT is a portable command-line driven interactive data and function plotting utility. GNUPLOT supports many types of plots in either 2D or 3D. It can draw using lines, points, boxes, contours, vector fields, surfaces, and various associated text. It also supports various specialized plot types.

Start the session with Insert→Session→Gnuplot; the program `gnuplot` (and PYTHON) must be installed, otherwise GNUPLOT is not in the menu. Type the commands of GNUPLOT; the session sends them to `gnuplot` with the terminal `postscript eps` and inserts the plot.

```
gnuplot] plot [-3.14:3.14] sin(x)
```




```
Gnuplot]
```

The plot is inserted at 0.8 of the width of the paragraph. A first line starting with `%` changes the size or the format of the picture of one input: for instance

```
% -width 0.5par -output svg
plot [-3.14:3.14] sin(x)
```

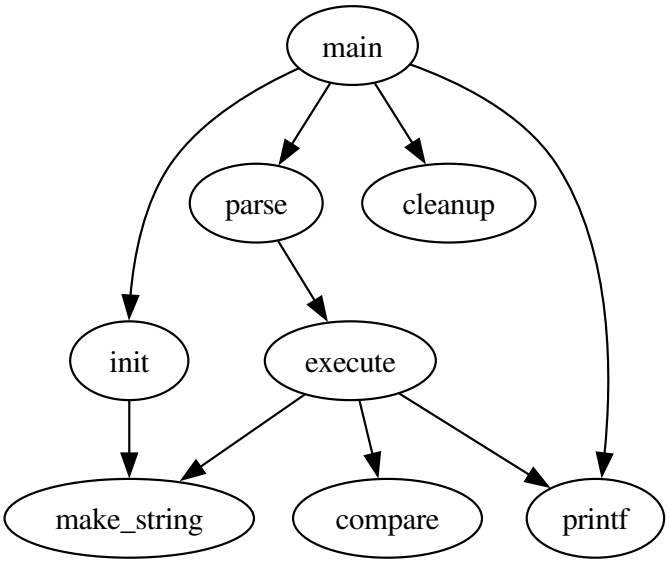
makes a smaller plot, in the SVG format (`eps`, the default, `png` or `svg`; the option `-height` also exists). The width is given in `par`, `pt` or `px`, the height in `pag`, `pt` or `px`; PNG and SVG pictures without a size come at their natural size. The options `-width` and `-height` also work in the sessions of DRATEX and XY-PIC, which always produce EPS; in the sessions of GRAPHVIZ, the line starts with the name of the layout program, as in `%dot -width 0.5par -output svg`.

4 GRAPHVIZ

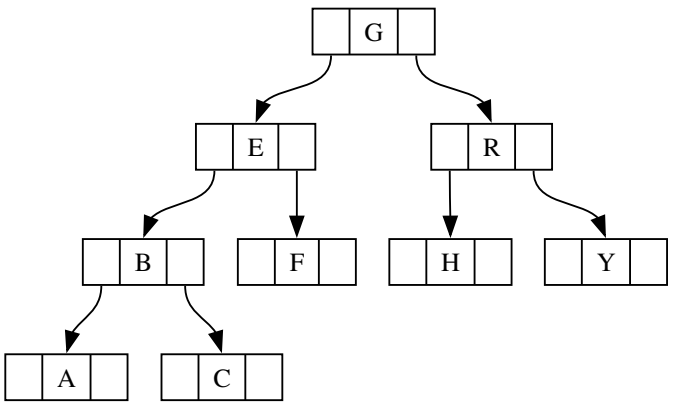
- Graph visualization is a way of representing structural information as diagrams of abstract graphs and networks.
- GRAPHVIZ is automatic graph visualization software. The GRAPHVIZ layout programs take descriptions of graphs in a simple text language, and make diagrams in several useful formats such as POSTSCRIPT for inclusion in TEXMACS.

Start the session with Insert→Session→Graphviz; the program `dot` of GRAPHVIZ must be installed. The session starts with the layout program `dot`; type `neato`, `twopi`, `circo`, `fdp`, `sfdp`, `patchwork` or `osage` alone in an input to switch to another layout program (and `help` for the list), or start one input with a line like `%neato` to use it only once.

```
   dot - graphviz version 16.1.0 (20260904.0139)
dot] digraph G {
       main -> parse -> execute;
       main -> init;
       main -> cleanup;
       execute -> make_string;
       execute -> printf;
       init -> make_string;
       main -> printf;
       execute -> compare;
     }
```



```
dot] digraph g {
       node [shape = record, height = 0.1];
       node0[label = "<f0>␣|␣<f1>␣G␣|␣<f2>␣"];
       node1[label = "<f0>␣|␣<f1>␣E␣|␣<f2>␣"];
       node2[label = "<f0>␣|␣<f1>␣B␣|␣<f2>␣"];
       node3[label = "<f0>␣|␣<f1>␣F␣|␣<f2>␣"];
       node4[label = "<f0>␣|␣<f1>␣R␣|␣<f2>␣"];
       node5[label = "<f0>␣|␣<f1>␣H␣|␣<f2>␣"];
       node6[label = "<f0>␣|␣<f1>␣Y␣|␣<f2>␣"];
       node7[label = "<f0>␣|␣<f1>␣A␣|␣<f2>␣"];
       node8[label = "<f0>␣|␣<f1>␣C␣|␣<f2>␣"];
       "node0":f2 -> "node4":f1;
       "node0":f0 -> "node1":f1;
       "node1":f0 -> "node2":f1;
       "node1":f2 -> "node3":f1;
       "node2":f2 -> "node8":f1;
       "node2":f0 -> "node7":f1;
       "node4":f2 -> "node6":f1;
       "node4":f0 -> "node5":f1;
    }
```




```
dot]
```

In the inputs of the session, the highlighting of the DOT language shows each space inside a string as a small open-box mark: type them as ordinary spaces.

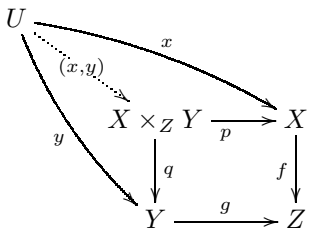
5 XY-PIC

- XY-PIC is a package for typesetting "matrix-like" diagrams, such as commutative diagrams. In category theory, "commutative diagrams" are the categorists' ways to illustrate equations and universal properties.

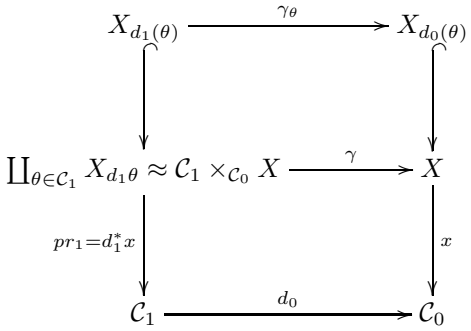
Start the session with Insert→Session→XYpic; it needs LaTeX with the package `xy` (it is in the usual TeX distributions, such as TeXLive). The input is put in a LaTeX document which loads `xy`, `amsmath` and `amssymb`.

```
TeXmacs interface to XYpic (high level 2-dimensional graphics)
```

```
xypic] \xymatrix{
        U \ar@/_/[ddr]_y \ar@/^/[drr]^x \ar@{.>}[dr]|-{(x,y)} \\
         & X \times_Z Y \ar[d]^q \ar[r]_p & X \ar[d]_f \\
         & Y \ar[r]^g & Z
       }
```




```
xypic] \xymatrix@!=2.5pc{
        X_{d_1(\theta)} \ar[rr]^-{\gamma_\theta}\ar@{^(->}[d] &&
        X_{d_0(\theta)}\ar@{^(->}[d] \\
        \coprod_{\theta\in {\mathcal C}_1}X_{d_1\theta}\approx
        {\mathcal C}_1\times_{{\mathcal C}_0}X
        \ar[rr]^-\gamma \ar[d]_{pr_1=d^*_1x} && X\ar[d]^x \\
        {\mathcal C}_1 \ar[rr]^{d_0} && {\mathcal C}_0
       }
```




```
xypic]
```

© Nicolas Ratier